

APPLIED RESEARCH

MonitorChain: An Extensible Tool for Real-Time Monitoring of Blockchain-Based Software Applications

ANDRÉ ARAÚJO^{ID}, (Member, IEEE), RENDRIKSON SOARES^{ID}, AND WALDSSON SANTOS^{ID}

Computing Institute, Federal University of Alagoas, Maceió 57072-970, Brazil

Corresponding author: André Araújo (andre.araujo@ic.ufal.br)

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ABSTRACT Support tools that monitor software application processes are required to identify, measure, and assess their performance and resolve any anomalies that impede their full functionality. Similarly, applications that use blockchain technology require analysis and monitoring tools for the resources and functionality provided by their smart contracts. This paper presents a computational tool that monitors hardware resources and evaluates transaction processing requests in blockchain networks. The solution presented here is called MonitorChain, and it has software components that allow the connection with different blockchain networks, the configuration of the workload, the monitoring of consumed hardware resources, and the evaluation of requests processed over blockchain networks. The effectiveness and adaptability of MonitorChain were validated through extensive evaluations across various blockchain environments, including public networks like Ethereum, Avalanche, and Fantom, as well as a private network using Hyperledger Fabric, with a consistent data load of 10,000 transactions. Additionally, through interviews with experienced software developers, we gathered feedback that led to further refinements and identified opportunities for future enhancements. The results demonstrated MonitorChain's capability to provide critical insights into blockchain applications' performance, scalability, and efficiency. MonitorChain advances the field compared to existing solutions by offering enhanced support for diverse networks, customizable workloads, and real-time data visualization, contributing significantly to the blockchain monitoring landscape.

INDEX TERMS Blockchain technology, software development tools, real-time monitoring tools, smart contracts.

I. INTRODUCTION

The software industry has created computer systems that use blockchain technology in several areas, including the Internet of things, streaming services, the health sector, and public safety [1], [2], [3]. Blockchain technology, viewed as disruptive, has transformed traditional business models due to its ability to provide security in data transactions, transparency and traceability in contracts, and cost reduction by eliminating a third party to mediate transactions between the stakeholders involved.

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Blockchain technology is used in the development of computer systems via decentralized applications (DApps). DApps are open source, run on a peer-to-peer network, and use smart contracts to interact with a blockchain network [4]. A key feature of a decentralized application is that it executes transactions autonomously and without the involvement of central mediators [5].

The software application development process using blockchain technology involves several key steps, beginning with the design and implementation of smart contracts, which are self-executing contracts with the terms of the agreement directly written into code. Developers then build the application to interact with these smart contracts, typically

using a backend system to manage data and business logic. REST APIs play a crucial role in this process by providing a standardized and simplified method for the application to communicate with the blockchain network. They allow developers to send transactions, query the blockchain, and retrieve data from smart contracts, enabling seamless integration and interaction between the software application and the blockchain. This approach enhances the scalability, interoperability, and efficiency of blockchain-based applications.

Software applications that utilize blockchain technology require significant computational resources and high Internet traffic availability. Additionally, the increased electricity consumption in transaction mining and the high carbon emissions during this process are highlighted as imminent challenges that require advancements in the current state of practice [6]. Recent literature research has focused on developing tools to evaluate the performance of both private and public blockchain networks [7]. In the state of the practice, solutions such as Hyperledger Benchmark [8], Distributed Ledger Performance Scan (DLPS) [9], and BlockBench [10] support different consensus protocols and offer custom testing and comprehensive performance measurement of blockchain applications.

Computational tools for this purpose enable the evaluation of processing performance and hardware resource consumption when a set of instructions is executed over a blockchain network. As a result, processing failures and bottlenecks can be identified, and architectural changes made to the evaluated software application.

Despite significant advancements in blockchain technology and the development of various tools for monitoring blockchain performance, there remains a notable gap in the state of the art regarding comprehensive monitoring tools specifically designed for software applications that use blockchain and interact with smart contracts via REST APIs. Existing research primarily evaluates performance metrics such as throughput, latency, and scalability [11], [12]. However, integrating these metrics with REST API interactions in a unified monitoring solution is still lacking, highlighting the need for tools for this purpose.

This paper presents a computational tool that monitors hardware resources and evaluates transaction processing requests in blockchain networks. The solution presented here is called MonitorChain, and it has software components that allow the connection with different blockchain networks, the configuration of the workload, the monitoring of consumed hardware resources, and the evaluation of requests processed over blockchain networks. In addition, MonitorChain provides additional graphical visualization resources for the functionalities described above. To evaluate the operation of MonitorChain, we developed a Rest API that used the components specified in the proposed solution and processed a set of transactions on both public and private blockchain networks to analyze the logs of hardware resource

metrics and client-server requests. The results indicated that MonitorChain could be used by other software applications, and the reports generated help understand the resources needed to run software systems on blockchain networks.

The main goal of our article is to support software applications that use blockchain technology and interact with smart contracts through REST APIs. REST APIs are crucial in blockchain application development, as they provide a standardized and simplified method for client-server communication. This technology, widely adopted in the software industry, enables seamless integration, improves interoperability, and reduces development time. A REST API monitoring tool for blockchain applications offers significant advantages, including efficient resource management, real-time performance evaluation, and streamlined troubleshooting. This ultimately improves the reliability and scalability of blockchain solutions.

The remaining sections of this article are organized as follows. Section II provides an overview of the foundational concepts and theories relevant to the research. Section III reviews the existing literature, highlights previous studies' contributions and limitations, and identifies the research gaps this work aims to address. Section IV introduces the proposed solution for monitoring and evaluating hardware resources and transaction processing requests in blockchain networks. Section V details the evaluation process used to assess the effectiveness of the proposed solution. Finally, Section VI summarizes the essential findings and discusses potential areas for future research.

II. BACKGROUND

This section provides an overview of blockchain technology, starting with its fundamental principles, decentralized architecture, and security mechanisms ensuring transparency and trust. It then explores public blockchain technology, explaining its open-access nature, consensus mechanisms, and diverse applications. Additionally, the section discusses private blockchain technology, emphasizing its restricted access, controlled participation, and suitability for enterprise environments that require enhanced privacy and data control. Finally, the section emphasizes the main concepts and importance of monitoring systems in software applications.

A. BLOCKCHAIN TECHNOLOGY

In recent years, blockchain technology has become one of the most impactful innovations on the technology scene [13]. Originating as the underlying infrastructure of the Bitcoin cryptocurrency, blockchain technology quickly spread beyond cryptocurrencies and found application in various areas, such as the Internet of Things (IoT), streaming services, healthcare, and public safety [14], [15]. A distinctive feature of blockchain technology is its ability to create decentralized, immutable, and transparent records of transactions, which are stored in a distributed network of nodes. Thus, blockchain

technology provides an additional layer of security and trust compared to traditional systems of records and transactions.

Blockchain is a distributed ledger technology based on a decentralized consensus system to create a shared, secure, immutable record of transactions or events [16]. This architecture allows information to be recorded in blocks linked sequentially in a chain and maintained by a decentralized network of validation nodes. Blockchain's underlying mechanism makes it resistant to tampering with records since any attempt to modify data requires the agreement of the majority of nodes in the network. This immutability and transparency of data and eliminating intermediaries in validation processes make blockchain a standout technology, with wide application in various areas, from finance to logistics, and the potential to profoundly transform recording and transaction practices across the entire spectrum of human activity.

B. PUBLIC BLOCKCHAIN NETWORK

Public blockchains are decentralized networks allowing individuals or entities to create, validate, and maintain transaction records. These networks are based on a consensus protocol, guaranteeing data integrity and security without a central authority [17]. Public blockchains, such as Bitcoin and Ethereum, operate in a transparent and immutable way, offering a secure platform for the execution of smart contracts and other decentralized applications (DApps). The fundamental characteristics of these networks include the transparency of transactions, the absence of a centralizing entity, the execution of smart contracts, and the ability to carry out peer-to-peer transactions without intermediaries.

In blockchain systems that use the Ethereum Virtual Machine (EVM) to execute smart contracts [18], the "Gas unit" represents the primary cost indicator in the network. Gas is an abstract measure that quantifies the computational effort required to carry out operations on smart contracts. This unit is crucial for understanding and regulating transactions and is paid in Ether based on the exchange rate called gas price.

An essential subdivision within Gas is the gwei, a smaller unit of Ether used to express Gas prices. 1 gwei is equivalent to 0.000000001 Ether, offering a finer granularity in transactions, especially useful in more complex smart contracts that require more detailed precision in measuring costs. In mining operations, miners have the autonomy to ignore transactions with gas prices considered too low, directly influencing the inclusion or exclusion of transactions in blocks. In the context of public blockchains, such as Ethereum, Gas is also used by miners to evaluate the performance and resource consumption of smart contracts, providing a fundamental metric for the economy and efficient operation of the blockchain network.

C. PRIVATE BLOCKCHAIN NETWORK

Private blockchains are permissioned networks that restrict participation to selected entities [19]. They offer centralized control over creating, validating, and maintaining transaction

records. Unlike public blockchains, private blockchains are designed for corporate environments where privacy and control are essential. Private blockchains like Hyperledger [20] Fabric allow for stricter participant access and permissions management, ensuring only authorized entities can interact with the network [21].

These networks make it possible to customize various aspects of their operation, such as the choice of consensus mechanisms and validation policies. The flexibility of private blockchains makes it possible to adapt operations and governance processes according to the specific needs of each organization. Data access control mechanisms can be implemented to ensure privacy and protect information between users.

Unlike public blockchains, storage on a private blockchain is not decentralized. Instead, the network operates a distributed ledger that functions as a restricted database, incorporating cryptography concepts, immutability, and organization-specific requirements [22]. Only authorized nodes can carry out transactions or validate changes to the ledger, ensuring access control over blockchain operations.

D. SOFTWARE APPLICATION MONITORING

Monitoring systems play a crucial role in analyzing and continuously improving the performance characteristics of computer systems. Monitoring systems are designed to collect, record, and analyze performance data and operational metrics from running applications [23]. These systems allow real-time visibility of the system's state, essential for identifying problems, performance bottlenecks, and anomalies that may arise during operation. In this sense, monitoring systems differ from benchmarks [24], which are developed as a set of metrics and tests designed to evaluate computer systems' performance in various scenarios and workloads. Thus, they are used in benchmarking environments based on static performance metrics. In contrast, monitoring systems are essential for the dynamic and real-time observation of system behavior in a production environment.

An application's continuous and high-quality operation is critical for meeting its users' needs. Support tools that monitor these applications' processes are required to identify, measure, and assess their performance and resolve any anomalies that impede their full functionality. Similarly, applications that use blockchain technology require analysis and monitoring tools for the resources and functionality provided by their smart contracts.

Blockchain networks, both public and private, enable organizations to manage their supply chains, contracts, payments, identities, and resources in novel ways. These blockchain-enabled changes in business models generate significant monetary value through daily transactions. In blockchain applications, tracking these transactions, ensuring their validity, and monitoring the consumption of computational resources to execute these processes is critical. To contribute to this vital research field, this paper

presents a computational tool called MonitorChain. This tool leverages REST API interactions to enable decentralized applications (DApps) to monitor hardware resources, evaluate client-server requests, and provide seamless interfaces for connections to different blockchain networks. Additionally, MonitorChain performs workloads to execute smart contract functionalities efficiently. By integrating these capabilities, MonitorChain offers a comprehensive solution for real-time performance evaluation, resource management, and monitoring in public and private blockchain environments.

III. RELATED WORK

This section presents a state-of-the-art analysis, comprehensively reviewing existing research and tools related to blockchain technology monitoring and evaluation. It examines various methodologies and frameworks developed for public and private blockchain networks. The discussion then focuses on identifying gaps in the current literature and highlights the main contributions of this paper.

A. STATE OF THE ART ANALYSIS

Monitoring and evaluating the performance of public blockchain networks is critical for understanding their scalability, efficiency, and overall effectiveness. Several studies have addressed these aspects, providing valuable insights and recommendations. For instance, in the work specified in [25], the authors introduce a thorough blockchain benchmarking framework (BBF) for public network scenarios in a private environment. The framework evaluates the network's performance across different configurations and workloads, measuring important metrics like throughput, latency, and resource utilization. Similarly, in [26], the study investigates how variations in consensus mechanisms affect performance metrics like transaction throughput and block propagation time, providing a comparative analysis of Ethereum's performance under different consensus algorithms. Additionally, another study offers an in-depth look at Bitcoin's performance limitations and scalability challenges, proposing solutions to improve Bitcoin's transaction processing times and overall network efficiency [27]. Furthermore, [28] evaluates the performance of three major public blockchain platforms: Ethereum, Hyperledger Fabric, and Bitcoin. The research highlights how different blockchain architectures impact performance by assessing their performance under varying workloads and configurations.

With the emergence of private blockchain networks for the needs of different business models, some research has focused on developing tools to evaluate software applications operating in these environments. In this field of research, the framework proposed in [10] is a comprehensive benchmarking that evaluates private blockchain platforms' performance, scalability, and security. It provides a detailed analysis of performance bottlenecks, identifying key areas where private blockchains can improve. The study sets a standard for benchmarking private blockchains and offers insights into the performance issues that affect these platforms.

Following a similar approach, the authors of [29] focus on the performance analysis of private blockchain platforms under varying workloads. The study highlights how performance metrics such as throughput, latency, and resource utilization change by testing different blockchain platforms with varying workload levels. The findings guide the selection and optimization of blockchain platforms to suit specific workload requirements, ensuring optimal performance in diverse scenarios.

Evaluating software applications on private blockchain networks is crucial to ensure optimal performance and security in business-critical environments. A study conducted in [30] carried out experimental performance evaluations across multiple private distributed ledger technologies. The analysis compared performance metrics such as latency, throughput, and scalability to understand how different ledger implementations perform under various conditions. This work helps to comprehend the performance trade-offs and make informed decisions about the most suitable ledger technology for enterprise solutions. Focusing on the Hyperledger Fabric blockchain platform, [31] identifies key performance bottlenecks and proposes optimization techniques to improve Hyperledger Fabric's performance significantly. These optimizations lead to better resource utilization and enhanced scalability, making the platform more efficient for enterprise applications. Likewise, an application-agnostic performance measurement framework for private blockchain platforms is proposed in [32]. The proposed framework standardizes how performance is measured and compared across different blockchain platforms, providing a comprehensive set of metrics for evaluating performance. The proposed solution aids developers and researchers in benchmarking performance and identifying areas for improvement, promoting a more uniform approach to performance analysis in the blockchain domain.

B. DISCUSSION

The related work section highlights significant contributions to blockchain performance evaluation and optimization. Studies [25] through [28] provide valuable insights into public blockchain networks' scalability, efficiency, and overall effectiveness. These studies present comprehensive evaluations of platforms like Ethereum and Bitcoin under various configurations, exploring the impact of different consensus mechanisms and identifying key performance metrics. They offer comparative analyses and propose solutions to address performance limitations, ultimately guiding the selection of the most suitable blockchain technologies for enterprise applications.

In the context of private blockchain networks, research has focused on developing tools to assess the performance of software applications in these environments. The framework proposed in [10] offers a detailed benchmarking of private blockchain platforms, evaluating their performance, scalability, and security and identifying performance bottlenecks. Similarly, [29] analyzes the performance of private

TABLE 1. Comparative analysis of related works.

Related Work		[10]	[25]	[26]	[27]	[28]	[29]	[30]	[31]	[32]
Metrics	Performance	●	●	●	●	●	●	○	●	●
	Scalability	●	●	○	●	●	○	●	●	●
	Security	●	○	○	○	○	○	○	○	○
	Latency	○	●	●	●	●	●	●	●	●
	Energy	○	○	○	○	○	○	○	○	●
Focus Area	Real time monitoring	○	○	○	○	○	○	○	○	○
	Benchmarking	●	●	●	○	●	●	●	●	●
Blockchain network	Private	●	○	○	○	●	●	●	●	●
	Public	○	●	●	●	●	○	○	○	○
Number of networks used	A network	○	●	●	●	○	○	○	●	○
	Two or more networks	●	○	○	○	●	●	●	○	●
Resources	Graphical user interface	○	○	○	○	○	○	○	○	○

blockchain platforms under varying workloads, providing insights that help optimize these platforms for specific workload requirements. Further contributions include studies [30], [31], [32], which evaluate multiple private distributed ledger technologies, propose optimization techniques for Hyperledger Fabric, and introduce an application-agnostic performance measurement framework. These works collectively enhance our understanding of blockchain performance trade-offs and promote a standardized approach to performance analysis in both public and private blockchain domains.

Table 1 compares the state-of-the-art works, highlighting the performance metrics assessed, application areas of the tools, and types of blockchain networks evaluated.

Our work distinguishes itself from existing research by focusing on monitoring and evaluating hardware resources and transaction processing requests in blockchain networks through a comprehensive tool called MonitorChain. While other studies, such as those evaluating Hyperledger Fabric, Ethereum, and Bitcoin, primarily assess performance metrics like throughput, latency, and scalability under various conditions, MonitorChain integrates REST API interactions with smart contracts. This integration provides a unified solution for real-time performance evaluation, resource management, and graphical visualization. Additionally, our solution offers support for private and public blockchain networks, catering to various setups. This holistic approach offers enhanced insights and practical applications for developers and researchers in the blockchain domain.

IV. PROPOSED SOLUTION

This section provides a broad overview of MonitorChain, explaining its objectives, capabilities, and software architecture. In addition, it highlights the web application interface, which facilitates real-time monitoring of blockchain-based software applications.

A. ARCHITECTURE AND OVERVIEW

MonitorChain is a computational tool that monitors hardware resources and evaluates transaction processing requests in public and private blockchain networks. Designed to allow the evaluation of different blockchain networks available

in the market, MonitorChain uses an architectural pattern that helps extend or develop new components without compromising the operation of the application.

Our tool is designed to provide real-time monitoring capabilities for blockchain-based software applications, offering a unique advantage through its integration with a REST API. This feature allows applications to continuously monitor client-server requests and hardware and software resources, ensuring optimal performance and resource management. The ability to monitor different blockchain networks in real-time through the REST API is the standout feature that differentiates our tool from other state-of-the-art solutions. By leveraging this capability, developers and administrators can gain immediate insights into network performance, resource utilization, and transaction processing, enabling more effective management and troubleshooting of blockchain environments. Figure 1 illustrates the software architecture of the proposed solution, and the definition of its main components is given below.

- **Rest API:** Plays a crucial role by enabling real-time monitoring and management of blockchain networks. It allows decentralized applications (DApps) to seamlessly connect with different blockchain networks, execute transactions, and retrieve performance data. The REST API's versatility and ease of integration make it essential for building scalable, efficient, and interoperable blockchain-based solutions.
- **Hardware Monitoring:** Tracks and calculates key metrics such as RAM usage, processing stack, and CPU utilization of the computing resource running the evaluation system. These metrics are updated every second, ensuring real-time accuracy, and the results are continuously recorded in a log file. This detailed logging provides an ongoing record of hardware performance, enabling users to monitor resource usage and optimize system efficiency.
- **Request Monitoring:** Designed to evaluate HTTP and HTTPS transactions, with monitoring metrics tailored to the specific blockchain network in use. This component records critical information in a log file for any type of network, including the request route address, HTTP or HTTPS response status codes, response time in

milliseconds, and the total number of successful and failed requests. In scenarios involving public Ethereum-based networks, additional metrics are captured, such as the number of gas units consumed in transactions, the cryptocurrency used, and the estimated cost of transactions in USD. This comprehensive monitoring ensures that all relevant data is tracked, providing valuable insights into network performance and transaction efficiency. Estimating the cost (EC) of blockchain transactions is determined by valuing the cryptocurrency used. This process involves taking into account the price of the gas unit in Gwei (GP), multiplied by the total gas used in the transaction (GE) associated with the selected digital currency, and multiplying it by the cryptocurrency's exchange rate in dollars (ER). The following equation explains the mathematical equation used for this calculation.

$$EC = \left(\left(\frac{GP}{1,000,000,000} \right) \times GE \right) \times ER$$

- **Connection interface:** To allow the connection with different blockchain networks, this component manages communication interfaces with the main DApps platforms available in the market was developed. The interfaces available in MonitorChain have connection parameters, persistence method signatures, and subscription contracts for business layer classes of software applications that need to communicate with different blockchain networks. MonitorChain already has communication interfaces with Fantom [33], Avalanche, [34], Ethereum [35] and hyperledger fabric networks [20], and if needed, new communication interfaces with other networks can be developed following the pattern implemented in the proposed architecture. However, in cases where the monitoring system is utilized in private networks that do not connect to public networks, the connection system through interfaces is not required.
- **Workload:** This component enables users to configure the amount of data processed across the available blockchain networks, offering flexibility to tailor the workload according to the specific needs of each operation. By adjusting the volume of data to align with the capacity and characteristics of different blockchain networks integrated with MonitorChain, users can optimize data processing for enhanced performance and efficiency. This adaptability ensures that MonitorChain can effectively meet the diverse demands of various blockchain environments, providing precise control over data handling and processing. Additionally, this component supports transaction creation, querying, and updating within blockchain transaction processing while enabling users to perform GET, POST, and PUT requests.
- **Hardware/Request Monitoring Log:** A detailed and structured record that captures all relevant data

regarding hardware performance and client-server requests within the blockchain network. This log is vital for real-time monitoring and historical analysis, supporting informed decision-making and proactive management of blockchain-based applications.

Our tool's extensibility is a fundamental aspect of its design, ensuring it can adapt and scale as needed to accommodate future developments and enhancements. The tool's architecture is modeled using the C4 model notation, which offers a clear and systematic representation of the software architecture at different levels of abstraction. Additionally, the architectural pattern is based on the REST principles, which guide the design of its components. These components communicate through well-defined interfaces, promoting low coupling and high cohesion within the system.

This modular design allows for easy expansion of the tool's capabilities. For instance, Ethereum-based blockchain networks can be added using the existing configurations within our tool, making integration straightforward. Additionally, new private blockchain networks can be incorporated by creating new interfaces in the source code, allowing the system to accommodate a broader range of blockchain environments without disrupting its core structure.

B. MONITORING WEB APPLICATION

To support real-time monitoring of software applications that utilize blockchain technology, we have developed a web-based application that enables users to visually and interactively monitor the critical features of our solution. Users can manage settings, track performance metrics, and receive alerts through an intuitive and user-friendly dashboard, all within a centralized platform. As illustrated in Figures 2 and 3, MonitorChain offers an intuitive graphical interface designed for monitoring software applications utilizing blockchain technology. Below is an overview of the key functionalities of the proposed solution.

The functionality of blockchain network settings within MonitorChain allows users to customize and manage the specific parameters of the blockchain environments they wish to monitor. This includes setting up network nodes, defining connection protocols, and selecting relevant blockchain networks, whether public or private.

The hardware monitoring functionality, as shown in Figure 2, provides users with real-time insights into the performance of the computing resources used by their blockchain applications. This feature includes visual panels that graphically display vital metrics such as central processing unit (CPU) usage, memory consumption, average stack usage, the percentage of CPU utilized, and the amount of memory used in megabytes (MB). Additionally, it estimates energy consumption, allowing users to gauge the power efficiency of their operations. MonitorChain provides a curated list of the main CPUs available on the market in the energy consumption estimation feature. Based on the technical specifications of each processor, the system

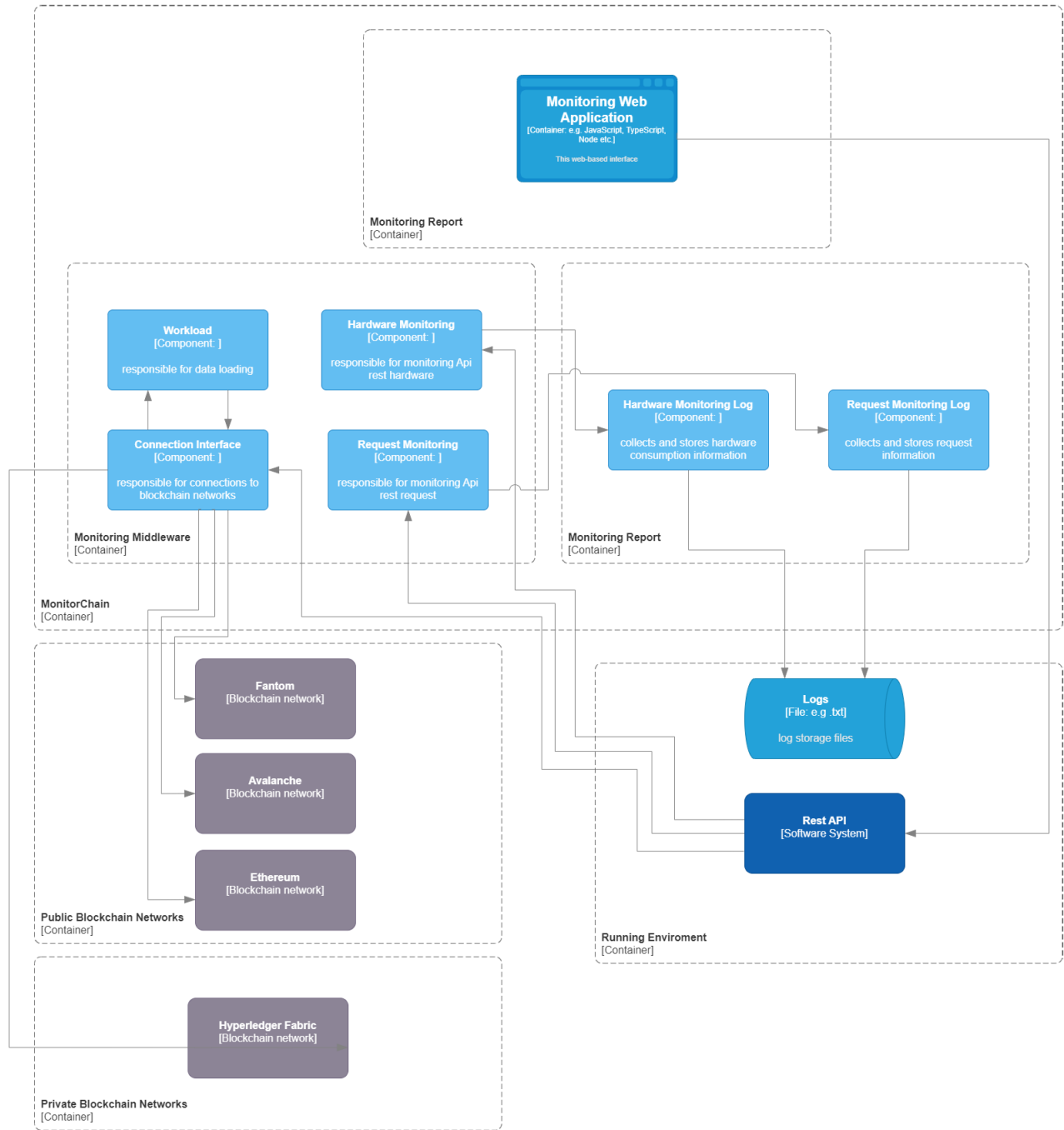


FIGURE 1. MonitorChain Software Architecture.

estimates the energy consumption in watts. This allows users to gain a more accurate understanding of the power usage associated with their hardware resources, enabling them to monitor and optimize the energy efficiency of their blockchain applications effectively.

The Request Information functionality, as illustrated in Figure 3, offers visual dashboards that provide detailed insights into various critical metrics related to blockchain transactions. These metrics include:

- **Total Requests:** This metric shows the cumulative number of client-server requests made within the monitored blockchain network.
- **Total Successful Requests:** This tracks the number of successfully processed and completed requests.
- **Total Failed Requests:** This indicates the number of requests that encountered errors or failed during processing, allowing users to identify and troubleshoot issues.

- **Total Gas Units Used:** For Ethereum-based networks, this metric records the total gas units consumed by transactions, which is crucial for understanding the resource cost of executing smart contracts.
- **Latest Method Request:** This shows the most recent method invoked in the blockchain network, providing insight into current operations.
- **Current Blockchain Currency:** This metric displays the cryptocurrency currently used within the network, which is essential for financial tracking.
- **Estimated Costs:** This estimates the financial cost of transactions based on the current blockchain currency and gas usage.
- **Response Time:** This tracks the time taken to process requests, measured in milliseconds, helping to assess the network's responsiveness and efficiency.

Finally, MonitorChain's workload functionality offers a highly customizable environment where users can configure the creation, update, and query transactions according to their needs. Users can define the number of requests to be sent, set the interval between each request, and specify the EndPoint address for the transactions.

V. EVALUATION

The evaluation section is structured to analyze MonitorChain's capabilities comprehensively. It begins with an overview of a software application scenario that leverages blockchain technology, serving as the foundation for evaluating MonitorChain's key functionalities. The subsequent sections present two distinct evaluations: the first examines the application's behavior on public blockchain networks, while the second assesses its performance within a private blockchain network. This dual approach facilitates a comparative analysis of MonitorChain's operation across blockchain environments. Furthermore, feedback from interviews with experienced software developers has informed refinements and identified opportunities for future enhancements. Finally, the section concludes with a comparative analysis of our solution, highlighting the main advancements and contributions of MonitorChain.

A. ASSESSMENT ENVIRONMENT

To evaluate our solution, we utilized a real-world scenario from a university setting and developed a software application that leverages blockchain technology to issue academic diplomas. This scenario was chosen to assess the practical applicability and effectiveness of MonitorChain in managing and securing critical academic processes. The university environment provided a robust test case for our solution, demonstrating how blockchain technology can be integrated into existing systems to enhance security, transparency, and efficiency in administrative operations. Figure 4 presents the flow of interactions in this scenario, and the main functional requirements are described below.

The defined scenario revolves around a blockchain-based academic diploma request system. When the student submits a diploma request, it is immediately registered in the system, automatically creating a smart contract containing the request's hash. The university then receives the request and the associated hash, which is used to verify the request's integrity. The system checks whether all academic conditions and requirements have been fulfilled. If the request meets the necessary criteria, the university confirms the completion of these requirements within the smart contract. Subsequently, the diploma is generated digitally, linked to the smart contract, and made available to the student. The smart contract maintains an immutable record of all transactions and states within the process, providing a transparent and secure trail of the entire diploma issuance process.

Based on the scenario described above, we integrated our solution, MonitorChain, to monitor the operation of the software application that utilizes blockchain technology for issuing academic diplomas. This integration allowed us to track and analyze the application's performance and resource usage in real-time, ensuring that each step of the diploma issuance process was efficiently managed and securely recorded.

B. EVALUATION OF PUBLIC BLOCKCHAIN NETWORKS

Our first round of evaluation focuses on monitoring the hardware resources and client-server requests in the university diploma request scenario, utilizing three public blockchain networks: Ethereum, Avalanche, and Fantom. This evaluation is designed to assess the application's performance and resource demands across these diverse blockchain environments, providing insights into how each network handles the operational workload involved in the diploma issuance process.

A robust computational setup was required to ensure accurate monitoring and performance analysis and to conduct this evaluation. The environment consisted of a high-performance server with the following specifications: Processor: Intel Core i9, CPU with eight cores and 16 threads, 32 GB of DDR4 RAM, a high-speed SSD with 1 TB of storage capacity, a high-bandwidth and stable internet connection, and operating System Windows Server, chosen for its reliability and compatibility with enterprise-level applications and blockchain tools.

For this evaluation, a data load of 10,000 transactions comprised 100 transactions for creating smart contracts, 6,000 for updating existing records, and 3,900 for querying information. The execution of these transactions was conducted individually on the three blockchain networks (i.e., Ethereum, Avalanche, and Fantom), allowing for a thorough comparison of how each network handles the specific workload associated with the diploma issuance process.

During the evaluation, it was possible to monitor the resources consumed during the execution of the data loads using the graphical interface provided by the

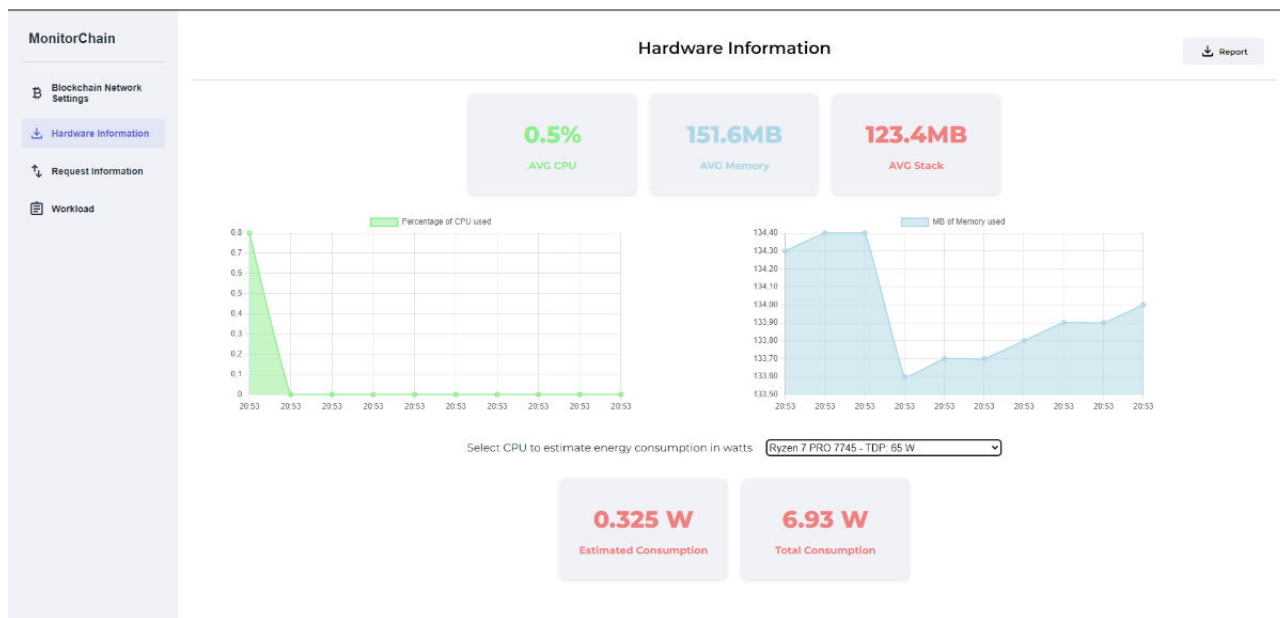


FIGURE 2. Hardware monitoring resources.

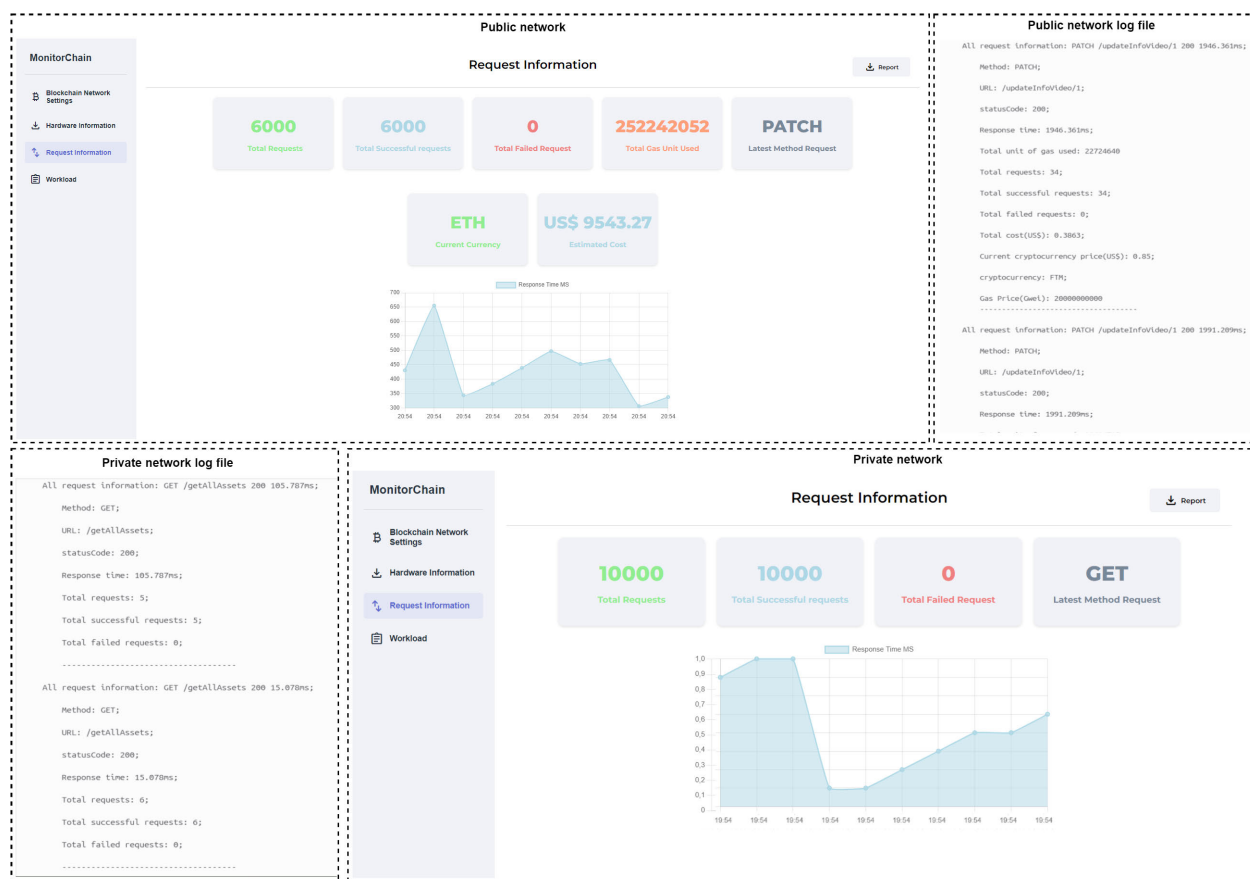


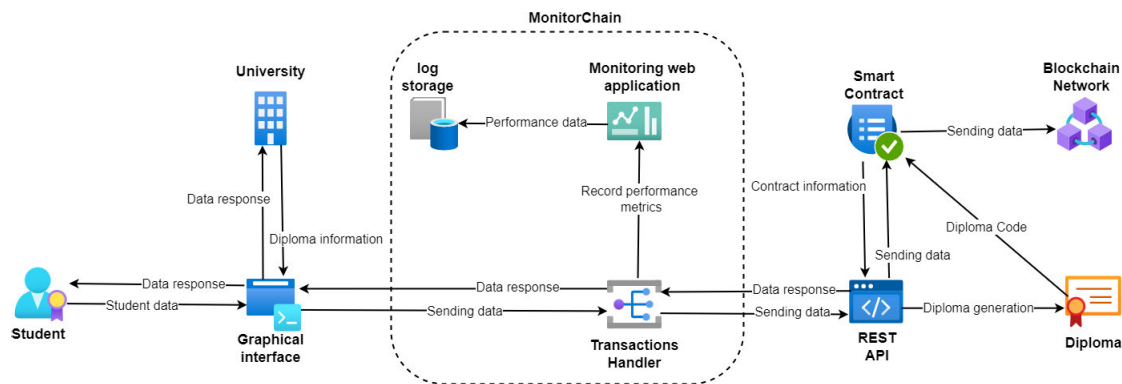
FIGURE 3. Request monitoring resources.

MonitorChain tool. As shown in Figure 3, the results obtained offer various information regarding the requests made at the end of this process. In the evaluation context, Figure 3

illustrates the results of the 6,000 transactions to update the student information in the contract linked to the diploma. This data covers not only the total number of requests but

TABLE 2. Execution costs for 10,000 transactions.

Network	Cryptocurrency	Market Price (US\$)	Gas Price (gwei)	Average Cost per Transaction (US\$)	Total Cost (US\$)
Ethereum	ETH	1891.490	20	00.97	9742.76
Fantom	FTM	00.32	3.5	0.0001	01.62
Avalanche	AVAX	20.71	3.5	0.010	102.94
Hyperledger Fabric	-	-	-	-	-

**FIGURE 4.** Blockchain application for applying for diplomas at a university.

also discriminates between successful and failed transactions, as well as shows the amount of gas used in all transactions, the cryptocurrency used, and the estimated costs related to the transactions.

In addition to the metrics already mentioned, it is worth noting that the collection of information on transaction costs also included a detailed analysis of the transaction fees associated with each operation and a description of the transactions according to their complexity and priority. This data provided a more comprehensive view of operating costs, enabling the identification of transactions with higher costs and providing input for strategies to optimize them. With this analysis, the evaluation can provide information for informed decision-making and continuous improvement of transaction processes in the monitored environment.

In the Ethereum network infrastructure, using the ETH cryptocurrency as a reference was based on a market price of US\$ 1891.49 per unit of ETH, while the gas price was set at 20 gwei. In this context, the average transaction processing cost was calculated at US\$ 00.97, resulting in a total cost of US\$ 9742.76. On the other hand, in the Fantom network, the reference cryptocurrency was FTM, whose market price was US\$ 00.32 per unit of FTM, and the gas price was set at 3.5 gwei. Under these conditions, the average cost of processing transactions was equivalent to US\$ 0.0001, resulting in a total cost of US\$ 01.62. Similar to the Ethereum network, the price of gas on the Fantom network played an essential role in determining operating costs and is vital to a comprehensive understanding of the financial aspects inherent in executing transactions. Finally, the total processing cost on the Avalanche network was US\$ 102.64. The data above on the total cost of executing 10,000 transactions on the three networks is summarized in

Table 2, providing a comparative view of the average costs per transaction and total costs on both platforms.

In addition, Figure 2 shows information on the hardware resources used throughout the evaluation process. This data includes average processor usage, RAM usage, and the processing stack. The graphs relevant to this stage illustrate the average use of processor capacity and the average amount of RAM used in the last 10 seconds, providing a more in-depth view of the performance of hardware resources during the execution of operations.

C. EVALUATION OF PRIVATE BLOCKCHAIN NETWORK

In the second round of evaluation, we monitored the operation of the diploma request application on the Hyperledger Fabric network. The evaluation was conducted using the same computational setup as in the first round, with the only difference being the operating system, which was switched to Linux to support the Hyperledger Fabric environment. The data load remained consistent, with 10,000 transactions being processed: 100 for contract creation, 6,000 for updates, and 3,900 for queries. This setup allowed us to effectively assess the application's performance on a private blockchain network and compare it to the results from the public blockchain networks evaluated earlier.

To conduct the tests and effectively utilize the monitoring tool, the API provided by Hyperledger Fabric was converted from a gRPC (gRPC Remote Procedure Call) architecture to a REST architecture. This conversion was necessary to enable the monitoring tool to access and interact with the data generated by user requests within organizations registered on the private network.

gRPC is an open-source framework designed for high-performance client-server communication using binary

serialization (Protocol Buffers). It supports bidirectional streaming and is well-suited for low-latency and high-throughput communication, making it ideal for microservices and complex distributed systems. However, due to its use of binary data and specific protocols, gRPC may not be as accessible for web-based applications and tools that rely on standard web protocols.

In contrast, REST is a more widely used architectural style that communicates over HTTP and uses standard text-based formats like JSON and XML. REST is language-agnostic and benefits from broad compatibility with web technologies, making integrating with various tools and applications, including our monitoring tool, easier. By converting the Hyperledger Fabric API to REST, we ensured that the monitoring tool could seamlessly interact with the blockchain network, accessing the necessary data to provide real-time insights and analysis.

Unlike public networks, monitoring data for private networks does not include information about gas consumption, currency used, or estimated transaction costs. This is because private networks have all nodes trusted and controlled by the organization itself, eliminating the need for these critical resources present in public networks to maintain consensus and manage transactions. Figure 3 shows the MonitorChain graphical interface, which displays request information for public and private networks and the logs generated during testing. This visual representation highlights the differences in monitoring metrics between these two blockchain environments, showcasing their distinct operational characteristics.

All transactions were executed on the private network, with the primary objective of this evaluation being not to conduct a comparative analysis of performance between public and private networks but rather to provide monitoring tools that enable real-time evaluation of hardware consumption characteristics and client-server requests. The focus is on ensuring that MonitorChain can effectively capture and display the relevant metrics critical for understanding the operational demands and performance of the application within a controlled, private blockchain environment.

D. INTERVIEW WITH IT PROFESSIONALS

We interviewed nine IT professionals with experience in blockchain technology to gather valuable feedback on our proposed software development tool.

Initially, we created a video explaining how the proposed solution works, covering the technology, the architectural model adopted, and the software components developed. Following this, we made the source code available in a GitHub repository and the entire environment used in the evaluations performed. The semi-structured interviews allowed the interviewees to express their opinions freely. This approach led to insights about our tool, providing us with critical feedback to refine and improve the solution. Table 3 displays the position of each interviewee in the IT market and their years of experience.

TABLE 3. Interview participants information.

Code	Job position	Work experience in years
P1	Software engineer	4
P2	IT Manager	18
P3	Software engineer	3
P4	Software engineer	12
P5	Software architect	18
P6	Administrator database	18
P7	Software engineer	6
P8	Software engineer	5
P9	Software architect	12

An important theme that emerged during the interviews was the recognition that in a real-world environment, monitoring software applications that use blockchain technology can be a costly activity, both in terms of resources and time. Three interviewees with extensive software industry experience suggested that an event-based communication approach could benefit the monitoring components. Specifically, they recommended using a messaging system to generate logs and handle monitoring tasks. This approach, they argued, would bring significant benefits to the execution of monitoring metrics, potentially reducing overhead and improving the efficiency and responsiveness of the monitoring process.

Another point that emerged during the interviews was the suggestion to create an alert generation service to notify users of unusual transactions that, according to predefined parameters, could compromise the application's execution. This proactive approach would allow quicker responses to potential issues, enhancing the system's reliability and security. In this context, two interviewees recommended implementing this functionality using microservices. A microservices architecture offers several advantages, including developing, deploying, and scaling individual services independently.

Continuing the analysis of the interview feedback, two interviewees with experience in cloud application software architecture suggested adopting an open standard called OpenTelemetry. OpenTelemetry is an open-source project that provides a set of APIs, libraries, agents, and instrumentation to generate, collect, and export telemetry data such as traces, metrics, and logs. It is designed to offer a unified standard for observability, enabling consistent data collection and monitoring across various platforms and services. According to the interviews, adopting this standard would allow us to leverage a broader ecosystem of tools and services that support this standard, potentially increasing the efficiency and effectiveness of our monitoring capabilities while ensuring that our solution remains flexible and scalable in diverse deployment scenarios.

Finally, the last theme that emerged from the interviews was the recognition of the significant potential of the proposed solution, MonitorChain, to be used in the software industry for monitoring applications and even supporting the viability of using blockchain technology in various business models, mainly by analyzing the financial cost metrics of transactions. Given this potential, it was suggested that MonitorChain be made available to the scientific and

TABLE 4. Analysis of monitorchain's key features.

MonitorChain		Properties
Metrics	Performance	●
	Scalability	●
	Security	○
	Latency	●
	Energy	●
Focus Area	Real time monitoring	●
	Benchmarking	○
Blockchain network	Private	●
	Public	●
Number of networks used	A network	○
	Two or more networks	●
Resources	Graphical user interface	●

developer community, allowing for the development and enhancement of its functionalities. Additionally, it was recommended that comprehensive documentation of the software artifacts, along with a detailed tutorial on how to use the tool, be created to facilitate adoption and further innovation within the community.

The interviews provided valuable insights and recommendations for enhancing MonitorChain, including adopting an event-based communication approach, implementing an alert generation service using microservices, and incorporating the OpenTelemetry standard. It was also suggested that MonitorChain be made available to the broader scientific and developer community with comprehensive documentation and tutorials. These interviews were crucial in identifying potential improvements and validating the practical applicability of our tool in real-world scenarios, ensuring its adaptability for future challenges in the software industry.

E. COMPARISON WITH THE STATE OF THE ART

In this section, we revisit the comparative table of the characteristics evaluated earlier in Section II. Now, after detailing all the functionalities of our solution, we present the advancements and features that MonitorChain offers. By re-examining Table 1 and considering the information provided throughout this document, we aim to highlight the significant improvements and capabilities that set MonitorChain apart from other state-of-the-art tools. This comparison underscores our solution's progress and innovative aspects, demonstrating how it addresses critical blockchain monitoring and management challenges.

As illustrated in Table 4, our solution provides a comprehensive set of hardware resource assessment metrics, except security metrics, as this area is not the primary focus of our proposal. As demonstrated throughout the proposal, we emphasize providing a real-time monitoring tool for software applications utilizing blockchain technology. This focus sets our solution apart from others primarily geared toward benchmarking tools.

One of the key differentiators of our solution is its support for both public and private blockchain networks, a feature demonstrated through our evaluations. This dual capability ensures that MonitorChain is versatile and applicable in

various blockchain environments, offering users flexibility and extensibility regardless of network configurations.

Additionally, our solution stands out for its robust monitoring environment, which is accessible through intuitive graphical user interfaces. These interfaces make the monitoring process more user-friendly and allow for the customization and processing of data loads according to specific needs. This combination of usability and flexibility ensures that users can effectively manage and optimize their blockchain applications in real-time.

VI. CONCLUSION AND THREATS TO VALIDITY

This paper presents MonitorChain, a computational tool designed to monitor hardware resources and evaluate transaction processing requests within blockchain networks. MonitorChain offers a range of software components that enable connections to various blockchain networks, workload configuration, monitoring of consumed hardware resources, and evaluation of processed requests. MonitorChain includes graphical visualization features to facilitate result interpretation, and logs can be downloaded after complete processing.

A key feature of MonitorChain is its ability to monitor blockchain applications across different networks. Its extensible and flexible architecture also allows it to support new networks. Additionally, MonitorChain provides a comprehensive set of metrics that assess blockchain-based software applications' technical and financial viability.

The evaluations conducted throughout this work have validated MonitorChain's effectiveness and adaptability as a powerful monitoring tool. We performed extensive assessments across various blockchain environments, including public networks such as Ethereum, Avalanche, and Fantom, as well as a private network using Hyperledger Fabric, all with a consistent data load of 10,000 transactions. Additionally, through interviews with experienced software developers, we gathered feedback that led to further refinements and identified opportunities for future enhancements.

Compared to the works identified in the state of the art, MonitorChain advances the field by providing extensive support for public and private blockchain networks, customizable workloads, and a user-friendly graphical interface for real-time data visualization. These achievements position MonitorChain as a significant contribution to the landscape of blockchain monitoring tools, with the potential to significantly influence the software industry's approach to blockchain technology.

Future work could investigate the integration of interoperability standards to improve communication between different blockchain networks within MonitorChain. Furthermore, exploring the adoption of the OpenTelemetry standard could standardize the collection and reporting of telemetry data across the monitored blockchain networks, ensuring consistent and reliable monitoring across diverse environments.

Although the results presented in this study demonstrate significant advancements in blockchain monitoring, it is essential to acknowledge the potential threats to validity that could influence the outcomes.

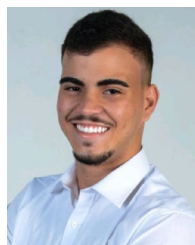
- **Construct validity:** If the programming routine or services used for data collection are flawed or inconsistent, this can compromise the validity of the measurements. For instance, if MonitorChain or any external monitoring tools have bugs, limitations, or inconsistencies in how they record data, the validity of the conclusions could be questioned.
- **Internal validity:** The assessment and results presented in this study are restricted to a specific scenario with few participants, features, and data transactions. Therefore, it is worth mentioning that the results indicated should not be generalized.
- **External validity:** Factors outside the scope of the study, such as variations in network conditions or hardware configurations, could influence the metrics used, potentially affecting the validity of the measured constructs.

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ANDRÉ ARAÚJO (Member, IEEE) received the Ph.D. degree from the Federal University of Pernambuco, Brazil, in 2018. He has an academic background in computer science. He is currently an Assistant Professor with the Federal University of Alagoas. With over a decade of experience in the software development industry, he was a Software Engineer and a Database Designer. He has authored several articles published in conferences and journals and leads research projects funded by government agencies focusing on digital transformation and data governance for the public sector. His research interests include software architecture, data modeling, data governance, blockchain, and software development methodologies.



WALDSSON SANTOS is currently pursuing the bachelor's degree in information systems with the Federal University of Alagoas. His research interests include software development, the IoT, blockchain, artificial intelligence, and innovation. He is focused on applying these technologies to create innovative solutions and expand his expertise in the field.

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RENDRIKSON SOARES is currently pursuing the master's degree in informatics with the Computing Institute, Federal University of Alagoas, Brazil. He has experience in blockchain, since 2018, with a focus on healthcare and streaming platforms. Currently, he is a Software Engineer with LCCV, engaging in naval and submarine engineering projects for the largest petroleum company in Latin America.